SkillUp

Lab: Generating BPMN, Process Flowcharts, Customer Journey Maps, and System Architecture Diagrams with Generative AI

Estimated time: 30 minutes

Learning objectives

After completing this lab, you will be able to:

- Generate BPMN diagrams from unstructured communications such as email threads
- Create flowcharts and customer journey maps from process descriptions
- Develop system architecture diagrams from high-level requirements

Introduction

In this lab, you will learn how to use AI to transform unstructured text into clear, professional diagrams. You will create BPMN diagrams, flowcharts, customer journey maps, and system architecture diagrams from real-world examples such as email threads, process descriptions, and system requirements. These visual models help you organize information, identify problems, and communicate effectively with stakeholders and development teams.

Required tools

- Web AI tool (ChatGPT or Claude)
- Diagramming tool that supports BPMN (Lucidchart, Draw.io, or Mermaid)

Exercise 1: Generate BPMN diagram from email thread

In this exercise, you will convert an email thread into a Business Process Model Notation (BPMN) diagram. BPMN diagrams show how tasks and decisions flow between people or systems, making complex processes easy to understand and analyze. You will identify actors and actions from the email thread, then use AI to generate the diagram structure.

Step 1: Extract actors and actions from the email thread

- Read through a sample email thread. Use a real email chain from your work or create a fictional one about a business process such as order fulfillment or customer support.
- Identify all actors, including people, teams, or systems mentioned in the thread.
- List the main actions or decisions each actor performs in the process.

Step 2: Generate the BPMN diagram using AI

- Open your AI tool and provide the email thread or your list of actors and actions.
- Prompt the AI tool to create a BPMN diagram showing the workflow between actors.
- Review the AI-generated output and verify all actors and key actions are included.

Hint

Example prompt:

```
"Based on this email thread, create a BPMN diagram in Mermaid syntax. Include swimlanes for each actor: [list actors]. Show the sequence of tasks, decision points, and handoffs between actors. Identify any bottlenecks or ineffi
ciencies in the process."
```

Sample actors and actions: Customer (submits order, provides payment info) → Sales Rep (reviews order, checks inventory) → Warehouse Team (picks items, packages order) → Shipping Dept (generates label, ships package) → Customer (receives order, confirms delivery)

Exercise 2: Create a flowchart and customer journey map from process description

In this exercise, you will generate two different types of diagrams from the same process description: a flowchart and a customer journey map. Flowcharts show the logical sequence of steps, whereas journey maps focus on the user's experience and emotions at each touchpoint. Both these diagrams reveal different insights about the same process.

Step 1: Analyze the process description

- Choose or write a process description. For example, how a customer signs up for a service, how a loan application is processed, or how a support ticket is resolved.
- Break down the description into sequential steps.
- Identify touchpoints where users interact with the system or make decisions.

Step 2: Generate both diagrams using AI

- Prompt the AI tool to create a flowchart showing the logical sequence of steps with decision points.
- Prompt the AI tool to create a customer journey map highlighting the user's experience, emotions, and pain points at each stage.
- Compare the two diagrams and note the different perspectives they provide.

Hint

Example prompt for flowchart:

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"Create a flowchart in Mermaid syntax for this process: [paste description]. Include start/end nodes, process steps, decision diamonds, and directional arrows showing the flow."
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Example prompt for journey map:

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"Create a customer journey map for this process: [paste description]. Include stages: Awareness, Consideration, Purchase, Use, Support. For each stage, identify: customer actions, touchpoints, emotions (positive/neutral/negativ
e), pain points, and opportunities for improvement."
```

Exercise 3: Generate a system architecture diagram from requirements

In this exercise, you will transform high-level system requirements into a system architecture diagram. Architecture diagrams show the major components of a system, how they connect, and how data flows between them. This visualization helps developers and stakeholders understand the technical structure before building anything.

Step 1: Identify the components from requirements

- Review a set of high-level requirements for a system. For example, an e-commerce platform, mobile banking app, or content management system.
- List the main components mentioned, such as databases, APIs, user interfaces, external services, and so on.
- Note how components connect and what data flows between them.

Step 2: Generate architecture diagram using AI

- Prompt the AI tool to create a system architecture diagram showing components and their connections.
- Ensure the diagram includes data flows, external integrations, and key technologies.
- Review the diagram for completeness and technical accuracy.

Hint

Example prompt:

```
"Create a system architecture diagram in Mermaid syntax for: [paste requirements]. Include these components: [list components such as Web App, Mobile App, API Gateway, Database, Payment Service, Authentication Service]. Show th
e data flow between components with labeled arrows. Indicate which connections use REST APIs, which use databases, and which are external integrations."
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Sample architecture: User → Web/Mobile App → API Gateway → [Business Logic Layer → Database] + [Payment Gateway] + [Auth Service]. Each arrow should show what data is exchanged (for example, 'user credentials', 'product data', 'payment token').

Summary

In this lab, you used AI to transform unstructured text into professional visual diagrams. You generated BPMN diagrams from email threads, created flowcharts and customer journey maps from process descriptions, and developed system architecture diagrams from requirements. These skills help you document processes, communicate complex information visually, and align teams more effectively.